

HDOC - Day 6 (C Programming)

C Programming Activity Report

Question 1 - Circular Garden

1. Problem Understanding, Input, Output and Formulae

Find the fencing cost and grassing cost of a circular garden when its radius, fencing rate per meter, and grassing rate per square meter are given.

Inputs: radius (m), fencing rate per meter, grassing rate per square meter.

Outputs: fencing cost and grassing cost.

Formulae:

Circumference = $2 \times \text{PI} \times \text{radius}$

Fencing Cost = Circumference x Fencing Rate

Area = $\text{PI} \times \text{radius} \times \text{radius}$

Grassing Cost = Area x Grassing Rate

Use PI = 3.14.

2. Standard Test Case Table

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	radius=5, fencing=10, grassing=20	Fencing=314.00; Grassing=1570.00	314.00; 1570.00	PASS
2	radius=10, fencing=5, grassing=10	Fencing=314.00; Grassing=3140.00	314.00; 3140.00	PASS
3	radius=2, fencing=15, grassing=25	Fencing=188.40; Grassing=314.00	188.40; 314.00	PASS
4	radius=0, fencing=10, grassing=20	Fencing=0.00; Grassing=0.00	0.00; 0.00	PASS

3. Algorithm and Evaluation

1. Start.
2. Declare radius, fencingRate, grassingRate, circumference, area, fencingCost, grassingCost.
3. Define PI = 3.14.
4. Input radius, fencingRate and grassingRate.
5. Calculate circumference = $2 \times \text{PI} \times \text{radius}$.
6. Calculate area = $\text{PI} \times \text{radius} \times \text{radius}$.
7. Calculate fencingCost = circumference x fencingRate.
8. Calculate grassingCost = area x grassingRate.

9. Display fencingCost and grassingCost.

10. Stop.

Evaluation: The algorithm calculates the boundary length for fencing and the circular area for grassing. The test cases cover normal values and the zero-radius case.

4. C Program

```
#include <stdio.h>

#define PI 3.14

int main()
{
    float radius, fencingRate, grassingRate;
    float circumference, area;
    float fencingCost, grassingCost;

    printf("Enter radius: ");
    scanf("%f", &radius);
    printf("Enter fencing rate per meter: ");
    scanf("%f", &fencingRate);
    printf("Enter grassing rate per square meter: ");
    scanf("%f", &grassingRate);

    circumference = 2 * PI * radius;
    area = PI * radius * radius;
    fencingCost = circumference * fencingRate;
    grassingCost = area * grassingRate;

    printf("Fencing Cost = %.2f\n", fencingCost);
    printf("Grassing Cost = %.2f\n", grassingCost);
    return 0;
}
```

5. Compilation, Execution and Test Evaluation

Compilation Command:

```
gcc garden.c -o garden
```

Execution Command:

```
./garden
```

6. Compilation and Execution

```
(kali㉿kali)-[~/hdoc6]
$ vi garden.c

(kali㉿kali)-[~/hdoc6]
$ gcc garden.c -o garden
```

```
(kali㉿kali)-[~/hdoc6]
$ ./garden
Enter radius: 5
Enter fencing rate per meter: 10
Enter grassing rate per square meter: 20
Fencing Cost = 314.00
Grassing Cost = 1570.00
```

Question 2 - Sector Area and Coloring Cost

1. Problem Understanding, Input, Output and Formulae

Find the sector area and coloring cost when the circular segment area, triangle area, and coloring rate per square centimeter are given.

Inputs: segment area (cm²), triangle area (cm²), coloring rate per cm².

Outputs: sector area and coloring cost.

Formulae:

Sector Area = Segment Area + Triangle Area

Coloring Cost = Sector Area x Coloring Rate

2. Standard Test Case Table

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	segment=50, triangle=30, rate=2	Sector=80.00; Cost=160.00	80.00; 160.00	PASS
2	segment=100, triangle=50, rate=5	Sector=150.00; Cost=750.00	150.00; 750.00	PASS
3	segment=25, triangle=15, rate=4	Sector=40.00; Cost=160.00	40.00; 160.00	PASS
4	segment=0, triangle=0, rate=10	Sector=0.00; Cost=0.00	0.00; 0.00	PASS

3. Algorithm and Evaluation

1. Start.
2. Declare segmentArea, triangleArea, coloringRate, sectorArea and coloringCost.
3. Input segmentArea, triangleArea and coloringRate.
4. Calculate sectorArea = segmentArea + triangleArea.
5. Calculate coloringCost = sectorArea x coloringRate.
6. Display sectorArea and coloringCost.
7. Stop.

Evaluation: The sector is the combination of the segment and its corresponding triangle. The coloring cost is calculated from the complete sector area.

4. C Program

```
#include <stdio.h>

int main()
{
    float segmentArea, triangleArea, coloringRate;
    float sectorArea, coloringCost;

    printf("Enter segment area: ");
    scanf("%f", &segmentArea);
    printf("Enter triangle area: ");
    scanf("%f", &triangleArea);
    printf("Enter coloring rate per square centimeter: ");
    scanf("%f", &coloringRate);

    sectorArea = segmentArea + triangleArea;
    coloringCost = sectorArea * coloringRate;

    printf("Sector Area = %.2f\n", sectorArea);
    printf("Coloring Cost = %.2f\n", coloringCost);
    return 0;
}
```

5. Compilation, Execution and Test Evaluation

Compilation Command:

```
gcc sector.c -o sector
```

Execution Command:

```
./sector
```

6. Compilation and Execution

```
(kali㉿kali)-[~/hdoc6]
└─$ vi sector.c

(kali㉿kali)-[~/hdoc6]
└─$ gcc sector.c -o sector
```

```
(kali㉿kali)-[~/hdoc6]
└─$ ./sector
Enter segment area: 50
Enter triangle area: 30
Enter coloring rate per square centimeter: 2
Sector Area = 80.00
Coloring Cost = 160.00
```

Conclusion

Both programs were designed using the required formulae, evaluated with standard test cases, and prepared for compilation and execution in Kali Linux.